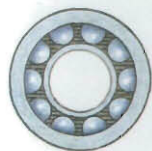


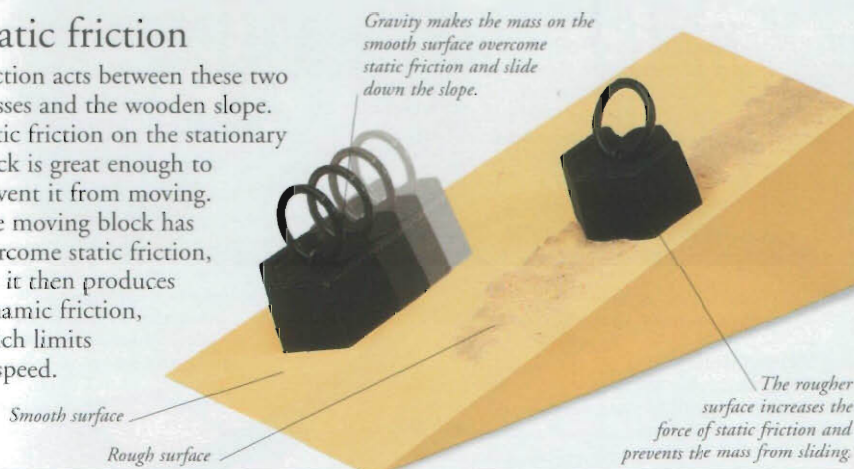
FRICTION



DRAGGING A HEAVY OBJECT across the floor is difficult because of friction, a force that opposes motion. Friction occurs between any two surfaces that are in contact, because even seemingly smooth surfaces have microscopic ridges and troughs that make them grip one another. Friction is greater between rough surfaces than smooth ones. Static friction stops surfaces at rest from moving. Dynamic friction slows down surfaces in motion.

Static friction

Friction acts between these two masses and the wooden slope. Static friction on the stationary block is great enough to prevent it from moving. The moving block has overcome static friction, but it then produces dynamic friction, which limits its speed.



Using friction

Friction can be useful. Without friction, no one would be able to walk or run. Friction helps people's shoes grip the ground and stops their feet from sliding out from under them. In the same way, friction enables a vehicle's tyres to grip the road. Most brakes use friction to slow a vehicle down. Friction between surfaces always produces heat, and sometimes electricity.



The pattern on the soles of these shoes is designed to create friction to give maximum grip.



Balloon charged by friction

Disc brakes

When the rider of this motorcycle applies the brakes, a pad presses against the metal disc fixed to the wheel. The rubbing action of the pad against the moving disc produces friction and slows the wheel enough to stop the motorcycle.



The brakes glow as the racing car slows down.



Friction and electricity

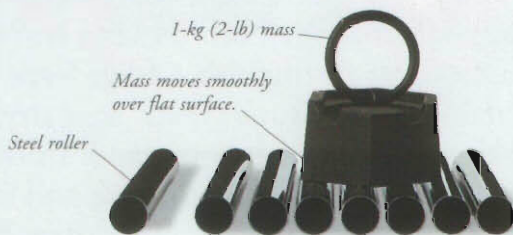
Rubbing two objects together can produce a charge of static electricity. Friction between a T-shirt and a balloon will dislodge negatively charged electrons from the atoms of the shirt. The electrons transfer to atoms in the balloon and give it a negative charge, while the shirt becomes positively charged. Opposite charges attract, so the negatively charged balloon clings to the positively charged shirt.

Friction and heat

In cold weather people often rub their hands together to warm them. Friction between two surfaces always produces heat. When a racing car brakes at high speed, the brakes glow red as the energy of the car's movement is changed into heat.

Reducing friction

If two moving machine parts rub together, friction will eventually damage them. Friction in machines generates heat and wastes a great deal of energy. Most methods of reducing friction involve keeping the surfaces apart in some way.



Rollers

The rollers under this 1-kg (2-lb) mass allow it to move smoothly over a flat surface. The mass and the surface are not in contact, so there is no friction between them. If the mass is pushed or pulled, the rollers will roll instead of dragging over the surface.



Ball bearing

A ball bearing is a device used to reduce friction between a wheel and its axle. As the wheel turns, the steel balls in the bearing roll around and prevent the wheel and axle from rubbing together.



Lubrication

Using a fluid such as oil to make machine parts move more smoothly and reduce wear is called lubrication. The oil coats the surfaces of the moving parts, preventing them from rubbing together.

Air resistance

Dynamic friction between the air and a moving object is called air resistance. Streamlining gives an object a smooth shape so that air flows more easily around it. Here, smoke trails blown over a streamlined car show how the air moves over its surface.

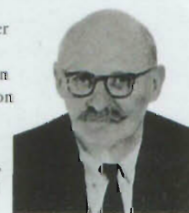


Hovercraft

A hovercraft overcomes the problem of friction by using high-pressure air as a lubricant. Fans pump air from the atmosphere into a flexible skirt around the craft's hull. This powerful downward jet of air allows the craft to hover over the water, reducing friction with the water's surface to a minimum. Large propellers on top of the craft move it quickly across the water.

Christopher Cockerell

In 1953, English engineer Christopher Cockerell (1910–99) began working on ways to reduce the friction between a ship's hull and the water. His solution was to lift the vessel above the water on a cushion of compressed air. In 1959, after making successful models, he produced the world's first practical hovercraft, called the SR-N1.



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